

APPLICATION FLOW DOCUMENT (AFD)

Carbon Horizon

Learn Your Impact. Shape the Future.

Version: 1.0

1. Application Overview

Carbon Horizon helps users understand, simulate, and reduce their carbon footprint through AI-powered insights, forecasting, and sustainability planning.

The application flow is designed to guide users from awareness to measurable carbon reduction.

2. High-Level User Flow

Landing Page

↓

User Registration/Login

↓

Onboarding

↓

Carbon Assessment

↓

Dashboard

↓

AI Sustainability Coach



Carbon Reduction Simulator



Carbon Horizon Forecast



Habit Tracker



Progress Monitoring

3. Detailed Application Flow

Phase 1: User Entry

Screen 1: Landing Page

Purpose:

Introduce Carbon Horizon.

Content:

- Project Vision
- Key Features
- Benefits
- Call-To-Action

Actions:

- Sign Up
- Login
- Learn More

Next:

User Registration

Phase 2: Authentication

Screen 2: Registration

Inputs:

- Name
- Email
- Password

System Actions:

- Validate inputs
- Hash password
- Create account

Output:

Account created successfully.

Next:

User Onboarding

Screen 3: Login

Inputs:

- Email
- Password

System Actions:

- Verify credentials
- Generate JWT Token

Output:

Authenticated session

Next:

Onboarding

Phase 3: User Onboarding

Screen 4: Profile Setup

Purpose:

Understand user lifestyle.

Inputs:

Age Group

- Student
- Professional
- Homemaker
- Retired

Transportation Preference

Diet Preference

Household Information

Sustainability Goals

Examples:

- Reduce carbon footprint
- Save electricity
- Reduce plastic usage

Output:

User Profile Created

Next:

Carbon Assessment

Phase 4: Carbon Footprint Assessment

Screen 5: Carbon Assessment Form

Purpose:

Collect lifestyle data.

Sections:

Transportation

- Car
- Motorcycle
- Bus
- Train
- Flight

Energy Usage

- Monthly electricity consumption
- AC usage

Food Habits

- Vegetarian
- Mixed
- Non-Vegetarian

Waste Management

- Recycling frequency
- Plastic usage

Action:

Calculate Footprint

System Process:

Input Data



Carbon Calculation Engine



Emission Factor Database



Emission Calculation

Output:

Carbon Footprint Result

Next:

Dashboard

Phase 5: Carbon Dashboard

Screen 6: Carbon Impact Dashboard

Purpose:

Visualize emissions.

Displays:

Total Annual Emissions

Example:

2.8 Tons CO₂e

Breakdown:

- Transportation
- Energy
- Food
- Waste

Charts:

- Pie Chart
- Bar Chart
- Trend Chart

Additional Insights:

Largest emission source identified.

Example:

"Transportation contributes 48% of your emissions."

Actions:

- View AI Suggestions
- Run Simulation
- Track Habits

Next:

AI Sustainability Coach

Phase 6: AI Sustainability Coach

Screen 7: AI Coach

Purpose:

Provide personalized guidance.

User Interaction:

Chat Interface

Example:

User:

"I travel 20 km daily by motorcycle."

AI:

"Switching to public transport twice a week could reduce your emissions by approximately 15%."

Capabilities:

- Sustainability Questions
- Carbon Reduction Tips
- Climate Education
- Personalized Recommendations

Actions:

Apply Recommendation

Next:

Phase 7: Carbon Reduction Simulator

Screen 8: Simulator

Purpose:

Test future lifestyle changes.

Available Simulations:

Transportation

- Car → Bus
- Motorcycle → Train

Energy

- Reduce AC Usage
- Reduce Electricity Consumption

Food

- Non-Veg → Mixed
- Mixed → Vegetarian

Waste

- Improve Recycling

System Process:

Current Footprint

↓

Apply Changes

↓

Recalculate Emissions

↓

Generate Comparison

Output:

Current Emissions

vs

Projected Emissions

Example:

Current:

3.2 Tons CO₂e

Projected:

2.4 Tons CO₂e

Reduction:

25%

Actions:

Save Scenario

Generate Forecast

Next:

Carbon Horizon Forecast

Phase 8: Carbon Horizon Forecast

Screen 9: Forecast Dashboard

Purpose:

Visualize future impact.

Scenarios:

Scenario A

Current Habits Continue

Scenario B

Partial Changes

Scenario C

Recommended Changes

Visualization:

Line Chart

Time Periods:

- 3 Months
- 6 Months
- 12 Months

Output Example:

Current Trend:

3.5 Tons CO₂e

Future Trend:

2.3 Tons CO₂e

Projected Reduction:

34%

Actions:

Create Action Plan

Next:

Habit Tracker

Phase 9: Habit Tracker

Screen 10: Sustainability Tracker

Purpose:

Track behavioral change.

Habits:

- Public Transport Used
- Recycled Waste
- Saved Electricity
- Conserved Water
- Reduced Plastic Usage

User Action:

Mark Daily Completion

System Actions:

Update Progress

Calculate Streaks

Estimate Carbon Savings

Outputs:

Weekly Report

Monthly Progress

Current Streak

Next:

Progress Monitoring

Phase 10: Progress Monitoring

Screen 11: Sustainability Progress Dashboard

Purpose:

Measure long-term impact.

Displays:

Initial Carbon Footprint

Current Carbon Footprint

Reduction Achieved

Forecast Improvements

Habit Completion Rate

Example:

Initial:

3.1 Tons CO₂e

Current:

2.5 Tons CO₂e

Reduction:

19.3%

Actions:

Run New Simulation

Update Assessment

Chat with AI Coach

4. System Data Flow

User Inputs



Frontend Forms



FastAPI Backend



Carbon Calculation Engine



Database Storage



Gemini AI Processing



Dashboard Generation



Visualization Layer



User Feedback

5. Error Handling Flow

Invalid Input



Validation Error



Display User-Friendly Message



Allow Correction

Gemini API Failure



Fallback Response



Display Basic Recommendations



Retry Option

Database Failure



Log Error



Notify User



Retry Request

6. Future Flow Enhancements

Version 2.0

- AI Bill Scanner
 - OCR-based Electricity Bill Analysis
 - Mobile Application
 - Community Carbon Challenges
 - Smart Device Integration
 - Personalized Carbon Budget
 - Gamification System
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7. Core User Journey Summary

Register



Complete Assessment



View Carbon Dashboard



Receive AI Recommendations



Run Carbon Simulations



Generate Future Forecast



Track Eco Habits



Reduce Carbon Footprint



Monitor Progress



Become More Sustainable